

# Topic: Data Control Language (DCL) Commands

**Q. What is Data Control Language (DCL)? Explain the concept of user privileges and demonstrate the use of GRANT and REVOKE commands.**

## 📌 Definition to Remember

**Data Control Language (DCL)** is a subset of SQL used by **Database Administrators (DBAs)** to control user access and maintain database security. It contains two commands: **GRANT** (give permissions) and **REVOKE** (remove permissions). DCL enforces the **Principle of Least Privilege**.

## 1. User Privileges and Access Control

A **Privilege** is a specific permission to perform an action on a database object.

| Privilege Type           | Description                      | Examples                          |
|--------------------------|----------------------------------|-----------------------------------|
| <b>System Privileges</b> | Administrative tasks             | CREATE TABLE , CREATE USER        |
| <b>Object Privileges</b> | Actions on specific tables/views | SELECT , INSERT , UPDATE , DELETE |

## 2. GRANT Command

Gives specific privileges to a user or role.

**Syntax:**

```
GRANT privilege_name ON object_name TO user_name [WITH GRANT OPTION];
```

- `WITH GRANT OPTION` allows the receiving user to further grant the privilege to others.

**Examples:**

```
-- Grant SELECT and INSERT on Employee table to user 'john'
GRANT SELECT, INSERT ON Employee TO john;

-- Grant all privileges to a manager role
GRANT ALL ON Employee TO manager_role;
```

## 3. REVOKE Command

Removes previously granted privileges from a user or role.

**Syntax:**

```
REVOKE privilege_name ON object_name FROM user_name;
```

**Examples:**

```
-- Remove INSERT from 'john' (he can still SELECT)
REVOKE INSERT ON Employee FROM john;

-- Remove all privileges from a former employee
REVOKE ALL ON Employee FROM ex_employee_user;
```

### ★ Must-Write Points (for 10 marks)

1. DCL controls **who can access** which database objects and **what actions** they can perform.
2. Two DCL commands: **GRANT** (give privileges) and **REVOKE** (remove privileges).
3. Two types of privileges: **System Privileges** (admin tasks) and **Object Privileges** (table actions).
4. **GRANT privilege ON object TO user** — grants specific access.
5. **WITH GRANT OPTION** allows the user to further pass the privilege to others.
6. **REVOKE privilege ON object FROM user** — removes a specific privilege.
7. DCL enforces the **Principle of Least Privilege** — give only the access needed.

### ⚡ Quick Recall

DCL → Security Control → Privileges (System + Object) → GRANT (give access, WITH GRANT OPTION) → REVOKE (remove access) → Principle of Least Privilege